

LUMABANK ACTIVATION EXPERIMENT

Project Summary

Scope · business context · data model

Fictional company · synthetic seeded data · generated 2026-09-11

00 — Scope & Scaffold (Phase 0)

Project: LumaBank — New-User Activation & the Onboarding Experiment That Broke Mid-Flight **Fictional company:** LumaBank — a Brazilian neobank (digital account + card). **Randomization:** per **user**, 50/50, via a feature-flag service. **Primary metric:** **Day-7 Activation** — account created **and** first transaction within 7 days. **Date locked:** 2026-09-10

PT — resumo. Projeto de portfólio nº 5, e o primeiro de **fintech**. A LumaBank é um neobank brasileiro fictício (conta digital + cartão), moeda **BRL apenas**. O time de Growth/Produto redesenhou o fluxo de **onboarding/KYC** (menos etapas, upload de documento adiado para *depois* da criação da conta) para aumentar a ativação de novos usuários. A métrica primária é **Day-7 Activation** (conta criada **e** primeira transação até o 7º dia). O experimento é **randomizado por usuário**, 50/50, com janela planejada de **6 semanas**. Por volta do **dia 11**, o monitoramento diário de **SRM (sample ratio mismatch)** detecta um desvio significativo da proporção 50/50 — que começa **exatamente no dia de um deploy do app mobile** sem relação com o experimento. A investigação precisa separar dois efeitos do bug (contaminação de quem já estava no teste × viés de atribuição dos novos cadastros, correlacionado com região/timezone), avaliar três alternativas de correção com trade-offs de viés × poder × prazo, e refazer o experimento numa **janela mais curta** com o **poder/MDE recalculado**. Não é a história de "rodei um A/B test e deu certo" — é a de um experimento que quebra em produção e é diagnosticado e refeito sob prazo.

1. Premise

LumaBank is a Brazilian neobank: users open a free digital checking account and a card entirely in-app. Growth is healthy at the top of the funnel — paid social, referral and app-store search all bring signups — but a large share of new users **never transact**. They download the app, start the sign-up, hit the KYC step (a full identity-document upload *before* the account exists), and drop. The users who do get through often take more than a week to make a first payment, by which point intent has cooled.

The Growth and Product teams redesigned the onboarding flow: **fewer steps to a working account, and the identity-document upload deferred to *after* account creation** (the account opens in a limited state, and the document is requested before limits are lifted or a physical card ships). The hypothesis is that getting a usable account into the user's hands sooner converts early intent into a first transaction.

This was launched as a **randomized experiment, unit = user, 50/50**, through the feature-flag service: on assignment a new user sees either the **new** onboarding (treatment) or the **existing** one (control). Planned window: **6 weeks**. Primary metric: **Day-7 Activation**.

What makes this project a portfolio piece is what happened next. Around **day 11**, the daily **sample-ratio-mismatch (SRM)** check — a standing part of how LumaBank runs any experiment — flagged that the arms were no longer 50/50, and the drift had started on the exact day of an unrelated **mobile app deploy**. The rest of the project is the incident: detection → root-cause diagnosis → a documented decision between three imperfect fixes under business time pressure → statistical re-planning → a clean re-run.

2. Locked scope decisions

Dimension	Decision
Deliverable language	English (all docs, code, reports). Every deliverable also carries a short PT explanation of each step. Working chat in Portuguese.
Priority	Phase 10 (data_analysis/) is the primary deliverable — the experiment incident, end to end. Power BI is secondary, scoped to 4 pages . Follows the AeroVanti model (analysis-primary, PBI lite), not VoltEdge/Voxa+/SwiftBite.
Portfolio gap this fills	No other project covers fintech , and none covers an experiment that breaks mid-flight on a real production bug and is diagnosed and re-run under deadline.
Geography / currency	Brazil only, BRL only . No second country, no FX — that ground is covered by VoltEdge.
Data window	21 months: 2025-01-01 → 2026-09-30 . Months 1–17 give the pre-experiment activation trend; the original experiment sits in month 19; the re-run finishes inside month 21 (with room for its Day-7 outcomes).
Original experiment window	6 weeks planned: 2026-07-06 → 2026-08-16 . Terminated early.
The break	Mobile app deploy on 2026-07-16 (experiment day 11) resets the local variant-assignment cache for a subset of users. Hotfix release 2026-07-22 .
Re-run window	4 weeks: 2026-08-24 → 2026-09-20 , bug fixed, power/MDE recomputed for the smaller sample the shorter window allows. The business could not wait another full 6 weeks.
Platform scale	~130,000 signups over the 21 months; end run-rate ~2,200–2,600 signups/week ; ~60,000 active users (≥1 transaction in the trailing 30 days) at window end.
Experiment scale	Original: ~13,500 randomized users (~6,750/arm). Re-run: ~9,000 (~4,500/arm).
ETL / data-quality depth	Balanced (matches the other four): 6 simulated source systems, 7 generic data-quality issue classes (\$5) plus a bug-specific catalog (\$6).
Power BI track	Semantic model as code (TMDL) + PBIR report + <code>theme.json</code> + build script; 4 pages , one business question per page.
Visual identity	New palette, designed in Phase 5 — must not reuse AeroVanti (navy/amber), VoltEdge (navy), or Voxa+ (violet/amber). Direction: trust/fintech without the money-green cliché.

3. Phase-1 report pages (4 pages, one question each)

#	Page	The question it answers
1	Executive Overview	Is new-user activation healthy and trending the right way, and what did the (re-run) onboarding experiment conclude?
2	Onboarding & Activation Funnel	Where do new users drop — signup → KYC → first transaction — and how does the funnel differ between the old flow and the new one?
3	Experiment Integrity Monitor	Can the experiment be trusted? Daily allocation by arm, SRM p-value over time, the deploy/break marked, contaminated vs clean population. <i>This page exists to tell the break story visually.</i>
4	Experiment Readout	Did the new onboarding work on the re-run — primary effect with CI, guardrails (KYC rejection / fraud) held, and does the effect differ by acquisition channel?

4. Experiment design (locked at Phase 0, detailed in Phase 1)

4.1 The original experiment (as pre-registered)

Element	Decision
Unit of randomization	User , at the moment they enter the onboarding flow (feature-flag exposure $\approx$ signup start).
Split	50/50 , treatment (new onboarding) vs control (existing onboarding).
Eligibility	Every new signup entering onboarding during the window, on a supported app version. Existing users are not re-onboarded and are out of frame.
Assignment	Feature-flag service hashes a stable user id $\rightarrow$ arm. Assignment timestamp and the serving app version are logged.
Treatment	Fewer onboarding steps; identity-document upload deferred to after account creation (account opens limited; document gates limit increases and the physical card).
Control	Current onboarding: full KYC document upload before the account is created.
Primary metric	Day-7 Activation rate = share of randomized users with account created and ≥ 1 transaction within 7 days of assignment.
Guardrail metrics	(1) KYC rejection rate — share of submissions rejected; must not worsen (non-negotiable — fintech). (2) Flagged-fraud rate — accounts with a fraud signal in the first 30 days; must not worsen (non-negotiable). (3) Onboarding support tickets per 1,000 signups — operational-cost guardrail. Health metric: time-to-first-transaction.
Planned analyses	SRM / balance check (see §4.3 — here it is continuous , not pre-flight only); stated hypothesis + MDE / power for the 6-week window ; primary effect with 95% CI; heterogeneity by acquisition channel with a formal arm $\times$ channel interaction test; novelty check across the window.
Planning values (to finalize in Phase 1)	Control Day-7 Activation $\approx$ 48% ; target detectable lift $\approx$ +3 pp ; $\alpha = 0.05$ two-sided, power = 80%; $\sim 6,750/\text{arm} \rightarrow \text{MDE} \approx$ 2.4–2.6 pp .

4.2 The break (deterministic, baked in Phase 3)

The 2026-07-16 mobile deploy reset the on-device variant-assignment cache for a subset of users. Two distinct effects, which the investigation must **separate**:

1. **Contamination** — users **already assigned before the deploy** whose cache was reset (`cache_reset_flag = 1`). A fraction of them received a **second assignment** on their next app open, sometimes to the **other arm** $\rightarrow$ they experienced part of one arm's treatment and part of the other's, mid-journey.
2. **Post-deploy attribution bias** — **new signups after the deploy** fell into an assignment **fallback** (`assignment_source = 'fallback'`) that does **not** randomize correctly: the fallback arm is a deterministic function of **region / timezone**. So the post-deploy arms differ **in composition** (geography, and everything correlated with it), not only in **proportion**. The fallback share decays as the 2026-07-22 hotfix rolls out.

4.3 SRM monitoring as a standing process (not a one-off)

Unlike AeroVanti — where the balance check is a single **pre-flight** test done once before trusting the result — LumaBank runs SRM monitoring **continuously for the life of every experiment**. Documented in Phase 1 as an operational **runbook**: a daily job runs a chi-square test on cumulative and trailing-7-day arm counts; an alert fires at **$p < 0.001$ confirmed on two consecutive days**; escalation pauses enrollment and notifies the experiment owner and platform on-call; the triage checklist checks recent releases, the `assignment_source` mix, and the per-region split. **This monitoring is what catches the bug** — it is the difference between this project and AeroVanti.

4.4 The decision (three options, documented with trade-offs)

Option	Bias	Statistical power	Timeline	Note
1. Truncate — use only data up to the day before the deploy	Low (clean randomization pre-deploy)	Very low — ~ 10 days, $\sim 1/4$ of planned N, and assignments in days 4–10 are right-censored before their Day-7 window closes	Immediate, but inconclusive	Honest but almost certainly can't clear the MDE
2. Exclude only the contaminated — keep clean pre-deploy users; drop arm-switchers; drop or re-weight the biased post-deploy fallback	Residual — can't prove every contaminated user was found; re-weighting rests on the observed covariates only	Moderate — loses $\sim 30\text{--}40\%$ of the post-deploy sample	A few days of analysis, no new data	Fastest defensible answer, but reviewer-fragile
3. Restart clean — fix the cache/fallback bug, re-randomize, run	None	Recomputed for whatever window the business allows	Worst — but capped at 4	The choice

Option	Bias	Statistical power	Timeline	Note
again			weeks, not another 6	

Chosen: restart, in a **4-week** window with the bug fixed and **power/MDE recomputed from scratch** for the ~9,000 users available — *not* the original 6-week MDE reused. The deep dive states the explicit cost of the deadline: at ~4,500/arm the re-run can only reliably detect a lift of roughly ≥ 3 pp, so a true effect in the **2–3 pp** range — which the original 6-week design was powered for — would now likely be missed.

4.5 The final readout (re-run, clean)

Hypothesis restated; power recomputed for the 4-week window; **primary effect** (Day-7 Activation, treatment – control) with 95% CI; **guardrails** — KYC rejection rate and flagged-fraud rate treated as **non-negotiable** (a faster activation bought at the cost of letting risk through is a fail), plus support-ticket load; **heterogeneity by acquisition channel** (paid social, organic, referral, app-store search, influencer) with a formal interaction test.

Underlying true effect: decided during Phase 3 (ETL) and baked into the generated data with plausible noise and channel heterogeneity. The true values live only in `etl/config.py` ; Phase 10 must detect and quantify them **as if blind**. The bug magnitudes (reset share, arm-switch fraction, fallback share and its region skew) are likewise set in Phase 3 and recovered by the investigation, not read off.

5. Data-quality issues to inject (generic set)

- 1. **Multiple date/time formats** across source exports — `DD/MM/YYYY` in the onboarding CSV, ISO-8601 in the KYC feed, Unix epoch-milliseconds in the transaction log.
- 2. **Timezone drift** — the risk/fraud system logs in **UTC**; everything else is `America/Sao_Paulo` , with no offset column.
- 3. **Money as strings** — `"R$ 1.234,56"` , `"1234.56"` , `"1.234,56"` mixed in transaction-amount and fee fields.
- 4. **Mojibake / mixed encoding** in user names and city fields.
- 5. **Duplicate rows** — the app retries on flaky mobile connections, so a slice of onboarding-step events and variant-assignment events are exact or near-exact duplicates.
- 6. **Missing master rows** — some `user_id` s appear in event feeds with no row in the user dimension export (referential gap).
- 7. **Snapshot-vs-log drift** — a "current variant" snapshot table that disagrees with the variant-assignment **event log**. (Diagnosticsly relevant: this disagreement is one of the first visible symptoms of the contamination.)

6. Bug-specific data catalog (the break, injected on purpose in Phase 3)

Element	Behaviour to simulate
<code>cache_reset_flag</code>	Emitted for ~12–18% of users already assigned before 2026-07-16 , on or shortly after the deploy.
Arm switch	~40% of the reset users get a second <code>variant_assignments</code> row on next app open; of those, ~50% land in the opposite arm → contaminated.
<code>assignment_source = 'fallback'</code>	For signups after 2026-07-16 : ~30–40% on day 1, decaying to ~0 over ~5 days as the 2026-07-22 hotfix rolls out.
Fallback bias	Fallback arm = deterministic hash of region / timezone , not the RNG → post-deploy arms diverge in geographic composition , not just in count.
<code>app_releases</code>	Contains the routine 2026-07-16 deploy (cause) and the 2026-07-22 hotfix (fix), among ~2–4 releases/month of noise.

7. Simulated source systems

System	Feeds	Grain
Feature-flag / experiment service	<code>variant_assignments</code> (user, arm, <code>assigned_ts</code> , <code>assignment_source</code> , <code>cache_reset_flag</code> , app version), <code>exposure_log</code> , <code>variant_snapshot</code> (current-state table)	assignment / exposure event
Onboarding / KYC service	<code>onboarding_steps</code> (step, ts), <code>kyc_submissions</code> (doc type, submitted_ts), <code>kyc_decisions</code> (approved / rejected / manual_review, reason code, decided_ts)	step / submission / decision

System	Feeds	Grain
Mobile app release log	<code>app_releases</code> (version, platform, <code>released_ts</code> , rollout %, notes) — the release calendar for root-cause cross-check	release
Core banking / ledger	<code>accounts</code> (user, opened_ts, status), <code>transactions</code> (account, ts, type, <code>amount_br1</code> , MCC) — first transaction = activation	account / transaction
Risk / fraud system	<code>fraud_signals</code> (user, flagged_ts UTC, signal type, score), <code>kyc_reason_codes</code> reference	fraud signal
CRM / marketing / support	<code>acquisition</code> (user, channel, campaign, first_touch_ts), <code>support_tickets</code> (user, opened_ts, category, channel)	user / ticket

8. Curated star schema (detailed in Phase 2)

- **Dimensions:** `dim_user`, `dim_date`, `dim_acquisition_channel` (paid social · organic · referral · app-store search · influencer), `dim_region` (UF + macro-region + timezone offset), `dim_experiment_arm` (control · treatment · ambiguous), `dim_app_release`, `dim_kyc_outcome`, `dim_onboarding_step`.
- **Facts:**
- `fact_variant_assignment` — **one row per assignment event** (contaminated users have ≥ 2): `user_key`, `arm_key`, `assigned_ts`, `assignment_source`, `cache_reset_flag`, `app_release_key`, `is_first_assignment`, `is_effective_assignment`.
- `fact_onboarding_step` — user × step × ts.
- `fact_kyc_decision` — one row per decision: `outcome_key`, `reason_code`, `is_rejected`.
- `fact_activation` — **one row per randomized user** (the analysis surface for the read-out): effective `arm_key`, `assigned_date_key`, `account_opened_flag`, `first_txn_ts`, `days_to_first_txn`, `day7_activated`, `cohort` (`pre_deploy` · `post_deploy` · `restart`), `contaminated_flag`, `in_analysis_flag`.
- `fact_transaction_day` — user × day (platform trend, time-to-first-transaction).
- `fact_fraud_event` — one row per fraud signal.
- `fact_support_ticket` — one row per ticket.
- `dq_checks.py` includes a check that **would catch the SRM as a data-quality failure** (arm-count chi-square by day), not only as an analysis step later.

9. Out of scope (explicit)

- A second country, region expansion, multi-currency / FX (covered by VoltEdge).
- Credit underwriting, limit-setting, lending — the account here is a checking account + card.
- A **fraud-detection model** — the flagged-fraud rate is a **guardrail input**, not modelled.
- Full customer-LTV / unit-economics modelling — activation is the outcome; downstream revenue is not modelled.
- Real-time / streaming data; everything is batch.
- App clickstream / screen-level telemetry beyond the onboarding-step events.
- Marketing-attribution modelling beyond a **first-touch channel** label.
- Real personal data — every user, account, transaction, KYC decision and fraud signal is synthetic and seed-generated.
- The **true treatment effect** and the **exact bug magnitudes** — set in Phase 3, detected blind in Phase 10.

10. Resolved-assumptions log

#	Assumption	Resolution
A1	Product	Free digital checking account + card. No credit/lending in frame.
A2	Window & anchor	21 months, 2025-01-01 → 2026-09-30; data-gen ("today") = 2026-09-10, so the last ~3 weeks are near-future relative to the build and exist only to close the re-run's Day-7 windows.
A3	Primary metric	Day-7 Activation = account created and ≥ 1 transaction within 7 days of assignment.
A4	"Active user"	≥ 1 transaction in the trailing 30 days (platform-health metric, not the experiment metric).
A5	Unit / split	User-level, 50/50, feature-flag hash assignment; assignment $\approx$ signup start.
A6	Original experiment window	6 weeks, 2026-07-06 → 2026-08-16; terminated early on the SRM alert.

#	Assumption	Resolution
A7	The break	Mobile deploy 2026-07-16 (day 11) resets the variant cache for a user subset; hotfix 2026-07-22.
A8	Two bug effects	(1) contamination of already-assigned users via arm switch; (2) region/timezone-correlated fallback bias for post-deploy signups. Baked in Phase 3, separated in Phase 10.
A9	SRM monitoring	Continuous daily chi-square (cumulative + trailing-7d); alert at $p < 0.001$ confirmed 2 consecutive days; documented as a runbook in Phase 1.
A10	Decision	Restart, not truncate or exclude-contaminated. Rationale + trade-off table in the deep dive.
A11	Re-run window	4 weeks, 2026-08-24 → 2026-09-20; ~9,000 users; power/MDE recomputed — original 6-week MDE not reused.
A12	Guardrails	KYC rejection rate + flagged-fraud rate = non-negotiable; onboarding support tickets / 1k signups = secondary.
A13	Heterogeneity	By acquisition channel, with a formal arm×channel interaction test.
A14	Planning values	Control Day-7 $\approx 48\%$; target lift $\approx +3$ pp; $\alpha = 0.05$, power = 80%. Finalized in Phase 1; true effect set in Phase 3.
A15	Currency	BRL only; no FX.
A16	True effect & bug magnitudes	Not decided here — set in ETL (Phase 3), detected blind in Phase 10.

11. Folder scaffold (created)

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LumaBank-Activation-Experiment/
docs/          00 scope · 01 business context & KPIs (+ SRM runbook) · 02 architecture ·
               03 data dict · 04 ETL · 05 model · 06 DAX · 07 visual identity ·
               08 data quality · 09 report structure · walkthrough.md · video script · PDF bundle
etl/           config (one SEED, true effect + bug magnitudes) · 01 generate raw ·
               02 clean/stage · 03 build curated · dq_checks (incl. SRM check) ·
               doc + DAX generators · run_pipeline · vocab/
data/
  raw/         source-faithful messy exports          (git-ignored, regenerated)
  staging/      typed stg_* tables                    (git-ignored, regenerated)
  curated/      Kimball star, CSV + Parquet            (git-ignored, regenerated)
  quality/      dq_report.md
powerbi/        semantic model as code (TMDL) · PBIR report · theme · build script
assets/         brand.py · logo / page-background generators
data_analysis/  sql/ python/ findings/ parity/ deliverables/ outputs/ ← PRIMARY
               (5-block investigation narrative: setup → SRM monitoring → root cause →
               decision analysis → re-run readout)
portfolio/      5 gallery images only
README.md

```

End of Phase 0 deliverable. Next gate: Phase 1 — `docs/01_business_context_kpis.md` (business context, KPI definitions, stakeholder personas, and the SRM monitoring runbook as a documented operational process). Review before Phase 2.

LumaBank — Business Context & KPI Framework

Document status: Phase 1 deliverable · authored at project kickoff, before any analysis. **Data generation date:** 2026-09-10 · **Platform-history window:** 2025-01-01 → 2026-09-30 (21 months — LumaBank's life to date). **Embedded experiment:** onboarding redesign A/B test — original 2026-07-06 → 2026-08-16 (terminated on day 11), clean re-run 2026-08-24 → 2026-09-20. **Audience:** LumaBank's Growth, Product, and Risk & Compliance leadership.

This document frames the problem and defines the metrics. It deliberately does **not** state whether the new onboarding worked, or by how much — that is the job of the read-out (Phase 10). Everything here was knowable *before* the analysis. The true underlying treatment effect, and the exact magnitudes of the production bug that broke the first run, are decided during ETL (Phase 3), baked into the synthetic data with plausible noise and channel heterogeneity, and kept only in `etl/config.py` ; Phase 10 must detect and quantify them **as if blind**.

PT — resumo. A LumaBank é um neobank brasileiro fictício (conta digital + cartão), lançado em jan/2025; a janela de 21 meses é a vida inteira da plataforma. O problema: muita gente se cadastra e **nunca transaciona** — o fluxo de onboarding/KYC (upload de documento *antes* de a conta existir) derruba os usuários. Growth/Produto redesenhou o fluxo (menos etapas, documento adiado para *depois* da criação da conta) e rodou um **A/B test randomizado por usuário, 50/50**, com métrica primária **Day-7 Activation** (conta criada e primeira transação até o 7º dia). Este documento define contexto, KPIs, personas e o **runbook de monitoramento de SRM** — o processo operacional diário que, no dia 11 do experimento, detectou a quebra.

1. The situation

LumaBank is a Brazilian neobank. A user downloads the app, opens a free digital checking account and gets a Mastercard — in principle in minutes, entirely in-app. The top of the funnel is healthy: paid social, referral, app-store search and influencer campaigns all deliver signups at acceptable cost. The problem is what happens next.

A large share of new users never transact. They install the app, begin sign-up, reach the **KYC step** — a full identity-document upload (RG or CNH, plus a selfie/liveness check) that today happens *before* the account is created — and drop. Of those who do get through, many take more than a week to make a first payment or transfer, by which point the initial intent that brought them in has cooled and a competitor's card is already in their wallet.

The Growth and Product teams' hypothesis: **the document-upload wall is in the wrong place**. A user who has to photograph two documents and pass a liveness check *before they have anything* is being asked to pay a cost before receiving any value. Move the account creation ahead of the document upload — open the account in a **limited state** (low transaction and balance caps, no physical card) and request the documents *before those limits are lifted* — and more users will reach a first transaction, faster.

In LumaBank's 19th month of operation the teams ran this as a **randomized, user-level A/B test**. The rest of this project is what happened to that test: it **broke in production on day 11**, was caught by the standing SRM monitor (\$9), diagnosed, and re-run clean in a shorter window (\$10–11).

2. Company profile

Name	LumaBank (legal entity <i>LumaBank Instituição de Pagamento S.A.</i>)
Type	Neobank / <i>instituição de pagamento</i> — digital checking account (<i>conta de pagamento</i>) + Mastercard debit; no credit product in scope
Headquarters	São Paulo, SP, Brazil
Launched	January 2025 (public launch). This project's 21-month window is the platform's whole life to date.
Market / currency	Brazil only, BRL only. No FX, no cross-border — that ground is covered by the VoltEdge project.
Reporting calendar	Calendar months; growth reporting weekly; experiment reporting daily
Scale at window end	~ 130,000 cumulative signups · ~ 60,000 active users (≥ 1 transaction in the trailing 30 days) · end-of-window signup run-rate ~ 2,200–2,600 / week
Regulatory frame	Bacen <i>instituição de pagamento</i> rules; KYC/AML obligations (identity verification, sanctions/PEP screening, transaction monitoring). KYC and fraud controls are a hard constraint , not a dial to trade away for growth — this is why they are non-negotiable guardrails

	(\$8.3).

3. The product and the two onboarding flows

3.1 What a user gets

A free digital checking account, a virtual Mastercard immediately and a physical card on request, Pix (instant payments), boletos, transfers, and a simple in-app statement. Revenue (out of scope to model) is interchange on card spend, float, and Pix-for-business fees. The **activation** the business cares about is the user's **first transaction** — Pix, card purchase, boleto payment or transfer — because a user who transacts once is an order of magnitude more likely to still be here in 90 days.

3.2 Control flow ("old" — document upload before the account)

open app → phone + CPF + email → basic personal data → ****upload RG/CNH + selfie/liveness****
→ KYC decision (approve / manual review / reject) → account created → first transaction

The KYC document step sits **before** the account exists. A rejected or stuck KYC means the user never had an account at all.

3.3 Treatment flow ("new" — account first, documents deferred)

open app → phone + CPF + email → basic personal data → ****account created (LIMITED state)****
→ first transaction possible within limits → prompt to upload RG/CNH + selfie
→ KYC decision → limits lifted / physical card enabled

Fewer steps to a usable account; the document upload is deferred and gates only **limit increases and the physical card**, not the account itself. Limited-state caps (indicative, finalized in Phase 3): balance ≤ R\$ 1,000, Pix ≤ R\$ 500/day, no boleto issuance, no physical card.

3.4 Why this is a genuinely two-sided bet

Moving the KYC wall later is expected to lift activation — but it also means **unverified users can transact**, briefly and within caps, before the platform has confirmed who they are. That is exactly the population where **KYC rejection** and **first-party / synthetic fraud** concentrate. The experiment therefore cannot be read on the primary metric alone; the risk guardrails (\$8.3) decide as much as the activation lift does.

4. The activation funnel and its definitions

Stage	Definition
Signup started	User submits phone + CPF and enters the onboarding flow. This is the randomization point — feature-flag exposure is logged here with a timestamp and the serving app version.
KYC submitted	Identity documents + liveness uploaded (in control, before account creation; in treatment, after).
KYC decision	approved · manual_review (routed to an analyst queue, later resolved to approved/rejected) · rejected (with a reason code).
Account created	A usable account exists (in control, only after KYC approval; in treatment, at the limited-state step).
First transaction	First Pix, card purchase, boleto payment or transfer of any amount. The activation event.
Day-7 Activation (primary metric)	Account created and ≥1 transaction within 7 calendar days of the randomization timestamp . A user assigned on 2026-07-06 12:40 has until 2026-07-13 12:40.
Day-1 / Day-30 Activation	Same construct at 1-day and 30-day horizons (context metrics).
Time-to-first-transaction (TTFT)	Hours from randomization to first transaction, for activated users. Reported as median and p90.
Active user	≥1 transaction in the trailing 30 days. Platform-health metric — not the experiment metric.

Indicative control-flow funnel, per 100 signups started (planning values; the real numbers come from the pre-period in Phase 10):

Signup started	KYC submitted	KYC approved	Account created	Day-7 Activation	Day-30 Activation
100	~82	~74	~74	~48	~58

The ~18-point fall from "signup started" to "KYC submitted" — the document-upload wall — is the drop the redesign is aimed at.

5. Acquisition channels

Randomization is independent of channel, but channel is the pre-registered **heterogeneity dimension** (§8.6): the redesign may help low-intent, top-of-funnel traffic (which bounces hardest at the document wall) more than warm, high-intent traffic — or the reverse. This is a prior to **test**, not to assume.

Channel	~Share of signups	Indicative control Day-7 Activation	Character
Paid social (Meta, TikTok)	~38%	~42%	Highest volume, lowest intent, most price/creative-driven
Organic / direct	~22%	~54%	Searched for LumaBank by name; warm
App-store search (ASO)	~18%	~50%	Category intent ("conta digital"), medium-warm
Referral (member-get-member)	~15%	~58%	Invited by an existing user; warmest, but capped volume
Influencer / affiliate	~7%	~40%	Spiky with campaigns; intent varies wildly by creator

`dim_acquisition_channel` also carries campaign and a first-touch timestamp; only the **first-touch channel** is modelled (no multi-touch attribution — out of scope, §16).

6. KYC, risk and fraud (guardrail context)

The controls that the experiment must not damage:

Control	Definition	Indicative baseline (control flow)
KYC rejection rate	Share of KYC submissions with a final decision of <code>rejected</code> (document invalid/illegible, identity mismatch, sanctions/PEP hit, suspected forgery).	~9% of submissions
KYC manual-review rate	Share routed to an analyst queue before resolution — an operational-cost signal.	~4% of submissions
Flagged-fraud rate	Share of activated accounts with a confirmed fraud signal (first-party default intent, synthetic identity, account-takeover, mule activity) within the first 30 days .	~1.2% of activated accounts
Chargeback / dispute rate	Card disputes per 1,000 card transactions in the first 30 days.	context only

Fraud signals come from the risk system (rules + model score + confirmed-loss backfill) and are timestamped in **UTC** while every other system is `America/Sao_Paulo` — a deliberate data-quality issue (§ architecture doc). The risk system is a **source of a guardrail metric**; this project does not build or tune a fraud model (§16).

7. The analytics window — why 21 months

Anchor	Value
Window length	21 complete months: month 1 = Jan 2025 , month 21 = Sep 2026
Pre-experiment baseline	months 1–17 (Jan 2025 – May 2026) — the activation trend before any test
Original experiment	month 19 → 2026-07-06 → 2026-08-16 (6 weeks planned; stopped day 11)
Clean re-run	months 20–21 → 2026-08-24 → 2026-09-20 (4 weeks)
Outcome tail	to 2026-09-30 — needed to close the Day-7 window for re-run users assigned as late as 2026-09-20 (outcomes through 2026-09-27)

Why 21 months. It is LumaBank's entire operating history — there is no earlier comparable data. The window also gives a long, clean pre-period so the experiment's control arm can be sense-checked against the platform's established activation rate.

Near-future caveat. The build ("today") is **2026-09-10**; the last ~3 weeks of the window are dated after it and exist only to let the re-run's Day-7 outcomes complete. No analysis treats them as observed history.

8. The onboarding experiment

8.1 Design (as pre-registered, before the break)

Element	Decision
Unit of randomization	User , at signup start. The feature-flag service hashes a stable pre-account user id → arm; the assignment and the serving app version are logged.
Split	50/50 , treatment (new flow, \$3.3) vs control (old flow, \$3.2).
Eligible population	Every new signup entering onboarding during the window, on a supported app version ($\geq$ the version that ships the new flow behind the flag). Existing users are never re-onboarded and are out of frame.
Assignment integrity	Set once at signup start; no re-randomization; no mid-flight flow changes for either arm. (<i>This is precisely the invariant the production bug violates.</i>)
Window	6 weeks: 2026-07-06 → 2026-08-16 . Chosen partly because July is a lower-volume winter-holiday stretch and 6 weeks were needed to accrue the target sample.
Expected sample	~ 13,500 users (~2,250/week) → ~ 6,750 per arm .

8.2 Hypotheses

- **Business hypothesis.** Creating the account *before* the document upload, and deferring KYC to a limits-gate, raises the share of new users who complete a first transaction within 7 days — **without** worsening KYC rejection or fraud, and without a material rise in onboarding support load.
- **Statistical hypothesis (primary).** H_0 : Day-7 Activation is equal in the two arms. H_1 : treatment > control. Tested one-sided at $\alpha = 0.05$; a two-sided 95% CI on the difference is always reported.

8.3 Metrics

Role	Metric	Definition	Decision rule
Primary	Day-7 Activation rate	\$4. Share of assigned users with account created and ≥ 1 transaction within 7 days of assignment.	Ship if the lift is positive, significant ($\alpha = 0.05$) and $\geq$ the MDE .
Guardrail — non-negotiable	KYC rejection rate	Rejected ÷ KYC submissions, by arm.	Non-inferiority: treatment must not exceed control by more than 1.5 pp (95% one-sided). A breach fails the experiment regardless of the activation lift.
Guardrail — non-negotiable	Flagged-fraud rate (first 30 days)	Confirmed fraud signals ÷ activated accounts, by arm. Uses only the fraud window observable within the data.	Non-inferiority: treatment must not exceed control by more than 0.5 pp (95% one-sided). A breach fails the experiment.
Guardrail — secondary	Onboarding support tickets per 1,000 signups	Onboarding-category tickets ÷ signups × 1,000, by arm.	Directional; flag if treatment is materially worse (operational cost).
Health / context	TTFT (median, p90) , Day-1 & Day-30 Activation, limited-state-never-verified share (treatment only)	\$4.	Context for whether a Day-7 lift is durable and verified, not just deferred paperwork.

8.4 Power & minimum detectable effect — the 6-week design

Assumption	Value
Control Day-7 Activation (planning value)	48%
Target effect (MDE), absolute	+3 pp (48% → 51%)
α (two-sided) / power	0.05 / 0.80

Assumption	Value
Required n per arm at the MDE (two-proportion)	≈ 4,450
Expected n per arm	≈ 6,750 → power ≈ 0.93 at +3 pp; ~0.80 power for an effect as small as +2.4 pp
Per-channel cells (per arm, expected)	paid social ~2,570 · organic ~1,485 · ASO ~1,215 · referral ~1,010 · influencer ~475 → well powered for paid social/organic, directional for referral/influencer

The power calculation is re-run from realized pre-period rates and reproduced in code in Phase 10; these are the planning basis. **These numbers do not carry over to the re-run** — §11 recomputes them for the shorter window and smaller sample.

8.5 Threats to validity the read-out must address

Threat	How it is handled
Sample-ratio mismatch (SRM)	Continuously monitored , not checked once — see the runbook, §9. χ^2 goodness-of-fit on arm counts, daily, overall and by segment. <i>This is the monitor that caught the break.</i>
Assignment integrity failure	The variant snapshot table is reconciled against the assignment event log every day; disagreements (arm switches) are surfaced as a data-quality failure, not silently resolved.
Covariate imbalance	Standardized mean differences on channel, region, app version, OS, signup hour, device age; Love plot. Especially important after the break, where the fallback biases composition.
Novelty effect	Daily treatment Day-7 Activation across the window; test for a decaying trend; report the settled lift.
Seasonality	Concurrent control arm differences out calendar effects; both arms' daily series still reported.
Interference between arms	Referral links can put an inviter and invitee in different arms; assumed second-order, stated as a bounded limitation.
Multiple comparisons	One primary metric fixed in advance; the two hard guardrails are one-sided NI tests; the channel heterogeneity cut is secondary and CI-reported with a family-wise-error note.
Peeking	The integrity monitor runs daily by design; the effect analysis is a single fixed read at window close. Daily SRM looks never estimate the treatment effect.

8.6 Heterogeneity analysis (required)

Effect on Day-7 Activation estimated **within each acquisition channel**, with per-channel CIs and a formal **arm × channel interaction test** (likelihood-ratio). The business decision — full rollout vs rollout scoped to specific channels — depends on this cut.

9. Runbook — continuous SRM & assignment-integrity monitoring

This is a **standing operational process**, documented here because it is part of how LumaBank runs *every* experiment — not a one-off analysis step. It is what detected the break on 2026-07-17. It is owned by the Experimentation Platform team; the on-call rotation carries the pager.

9.1 Purpose

Detect, within one day, any failure of the randomization to deliver the intended allocation (SRM) or the intended **stable** per-user assignment (integrity), for the full life of a running experiment.

9.2 Cadence & inputs

- Runs **daily at 07:00 BRT** as a batch job, for every experiment in `running` state.
- Reads: the prior day's `variant_assignments` events, the cumulative assignment log, the `variant_snapshot` current-state table, `app_releases` , and a backend deploy/flag-config changelog.

9.3 Tests

#	Test	Grain
T1	Pearson χ^2 goodness-of-fit vs expected 50/50	cumulative assignments to date
T2	Pearson χ^2 vs 50/50	trailing-7-day assignments

#	Test	Grain
T3	χ^2 vs 50/50, broken down	by <code>assignment_source</code> (normal / fallback), region, timezone, app version, OS
T4	Snapshot-vs-log reconciliation: count users whose current snapshot arm $\neq$ their first logged assignment arm	per user, aggregated
T5	First-assignment vs effective-assignment drift: count users with ≥ 2 assignment events	per user, aggregated

9.4 Alert logic

- **WARN** — T1 or T2 $p < 0.01$ on a single day, or T4/T5 count above a small fixed threshold. Logged, digested to the experiment owner; no pause.
- **ALERT** — T1 $p < 0.001$, or T2 $p < 0.001$ on two consecutive days, or T4/T5 rising day-over-day. Thresholds are deliberately tighter than 0.05 because the test runs every day for weeks (a naïve 0.05 would false-alarm on most experiments).

9.5 On ALERT

1. **Auto-pause new enrollment** into the experiment (existing assignments keep serving).
2. **Page** the experiment owner and the Experimentation Platform on-call.
3. **Freeze** an analysis snapshot as of the last day that passed all tests ("last clean day").
4. Open an incident doc; begin triage (§9.6).

9.6 Triage checklist (in order)

1. **Release diff.** Any app release, backend deploy, or flag-config change in the 48 h before drift onset? Cross-reference `app_releases` and the deploy changelog. (*In this incident: the 2026-07-16 mobile release.*)
2. **Source split.** Break SRM by `assignment_source`. Is `fallback` non-zero or rising? When did the first fallback assignment appear?
3. **Segment split.** Is the imbalance concentrated by region / timezone / app version / OS? A geographic concentration points to a biased fallback, not random drift.
4. **Integrity.** Quantify T4/T5 — how many users have switched arms, and were they assigned before or after the onset?
5. **Population partition.** Count users assigned **before** onset (candidate-clean, but check for cache-reset), users assigned **after** onset (candidate-biased), and the overlap that experienced a reset.

9.7 Outputs the runbook hands to the decision

Contaminated-vs-clean user counts with an auditable rule; number of usable clean days and the sample they contain; whether the defect is **still active** or already fixed by a hotfix; the covariate-imbalance profile of the post-onset population. These feed directly into the three-option decision analysis (§ scope doc §4.4, and Phase 10 block 4).

10. The break (summary — full treatment in Phase 10)

On **2026-07-16** (experiment day 11) a routine mobile app release reset the on-device variant-assignment cache for a subset of users. The 07:00 run on **2026-07-17** raised a WARN on T2; **2026-07-18** confirmed it to an ALERT and enrollment auto-paused. Two distinct effects, which the investigation must separate:

1. **Contamination** — users **already assigned before 2026-07-16** whose cache was reset (`cache_reset_flag = 1`); a fraction were re-assigned on next app open, some to the **opposite arm**, so they experienced a mix of both flows mid-journey.
2. **Post-deploy attribution bias** — signups **after 2026-07-16** fell into an assignment **fallback** (`assignment_source = 'fallback'`) whose arm is a deterministic function of **region / timezone**, not random. The post-deploy arms differ in **composition**, not only in count. The fallback share decays as the **2026-07-22 hotfix** rolls out.

The magnitudes (reset share, arm-switch fraction, fallback share and its geographic skew) are set in Phase 3 and recovered by the analysis, not read off.

11. The re-run

Element	Decision
Preconditions	Cache-reset handling fixed (a reset no longer triggers re-assignment); the biased fallback removed (fallback, if ever hit, now uses the same hash as the primary path). Verified in staging before restart.

Element	Decision
Fresh randomization	All users re-hashed; no carry-over of arms from the aborted run; aborted-run users entering again are re-randomized and flagged.
Window	4 weeks: 2026-08-24 → 2026-09-20. The business could not wait another full 6 weeks; 4 weeks was the agreed cap.
Expected sample	~ 9,000 users (~4,500/arm) — a slightly higher late-window run-rate plus modest pent-up demand.
Power / MDE — recomputed	At ~4,500/arm, $\alpha = 0.05$, power = 0.80 → MDE $\approx$ 3.0 pp (vs 2.4 pp for the 6-week design).
Stated cost of the deadline	A true effect in the +2 to +3 pp range — a plausible real size for this change, and detectable under the original design — will now likely come back non-significant . The deep dive states this explicitly rather than treating a null as "no effect".
Guardrails & heterogeneity	Unchanged from \$8.3 / \$8.6, re-powered for the smaller sample (guardrail NI tests lose power too — noted).

12. Seasonality and patterns embedded in the data

Pattern	Where it shows
January "organize my finances" surge	Signup volume peaks; paid-social CAC falls
Carnaval (Feb)	Signups and support contacts dip for ~1 week
Feb–Mar back-to-school	Secondary signup lift
Black Friday / November	Card-acquisition marketing push → signup spike; card-transaction spike
December 13th-salary	Transaction volume and first-transaction rate rise; more Day-7 activations per signup
Jun–Jul winter holidays	Lower signup volume — the original experiment sits here, which is why 6 weeks were planned
Late Aug–Sep	Volume recovers — the re-run runs into a rising-volume stretch
Paid-social volume	Tracks quarter-end marketing pushes; creative refreshes cause week-to-week swings
App releases	~2–4 per platform per month; the 2026-07-16 release was routine and not flagged as risky at the time
Weekly cycle	Signups skew to evenings and Sun–Tue; first transactions cluster on the assignment day and the following weekend

13. Stakeholder personas → report pages

Four personas, one standing question each — those four questions are the Phase-8 report pages (the dashboard is the secondary deliverable, kept lean).

#	Persona	Standing question	Page
1	VP Growth / GM	Is new-user activation healthy and trending up, and did the onboarding redesign work well enough to roll out?	Executive Overview
2	Product Lead, Onboarding	Where exactly do new users drop between signup, KYC and first transaction — and how does the new flow reshape that funnel?	Onboarding & Activation Funnel
3	Experimentation / Data Science Lead	Can the experiment be trusted — allocation over time, SRM, contamination — and once it is clean, what is the effect?	Experiment Integrity Monitor
4	Head of Risk & Compliance	Did faster activation cost anything on KYC rejection or fraud? Are the guardrails intact?	Experiment Readout (guardrail section)

14. KPI framework

Benchmark ranges are **directional**, drawn from public Brazilian-fintech and neobank commentary and payments-industry notes, adapted to a young single-country neobank. They set targets; they are not claims about named competitors.

14.1 Platform health & scale

KPI	Definition	Benchmark	Target
Cumulative signups	Distinct users who ever started onboarding	—	on plan curve
Active users	≥ 1 transaction in trailing 30 days	—	≥ 60k by window end
Active rate %	Active ÷ (accounts created)	45%–65%	≥ 55%
New signups / week	Signups starting in the week	—	≥ plan
Account-creation rate %	Accounts created ÷ signups started	60%–80%	≥ 75%
MAU / signup cohort retention	Share of a weekly signup cohort active at week n	M1 ≥ 55%	context

14.2 Acquisition

KPI	Definition	Benchmark	Target
Signups by channel	Count by first-touch channel	—	mix on plan
Channel mix %	Channel share of signups	~38/22/18/15/7	context
Day-7 Activation by channel	Primary metric split by channel	40%–58%	context / heterogeneity input
Blended Day-7 Activation	All channels, all signups	45%–52%	≥ 50%

14.3 Onboarding funnel

KPI	Definition	Benchmark	Target
Step conversion	Share advancing signup → KYC submit → KYC decision → account → first txn	—	raise the KYC-submit step
KYC-submit rate %	KYC submitted ÷ signups started	70%–88%	≥ 85% (the redesign's job)
Account-creation rate %	§14.1	60%–80%	≥ 75%
Median time in onboarding	Signup start → account created (minutes)	4–15	≤ 8
Limited-state-never-verified % (<i>treatment only</i>)	Accounts still limited (no KYC) 14 days after creation	—	≤ 15%

14.4 Activation

KPI	Definition	Benchmark	Target
Day-7 Activation rate % (primary)	§4	40%–55%	≥ 52%
Day-1 / Day-30 Activation rate %	§4	—	context
Median / p90 TTFT	Hours from assignment to first transaction (activated users)	median 12–48 h	↓
First-transaction type mix	Pix / card / boleto / transfer share of first transactions	Pix-led	context
Activation by signup week	Day-7 Activation of each weekly cohort	—	trend up

14.5 KYC & risk — guardrails

KPI	Definition	Benchmark	Target
KYC rejection rate %	Rejected ÷ KYC submissions	5%–12%	≤ 10% and not worsened by the experiment
KYC manual-review rate %	Manual-review ÷ submissions	3%–8%	≤ 6%
Flagged-fraud rate % (30d)	Confirmed fraud signals ÷ activated accounts, first 30 days	0.5%–2.0%	≤ 1.5% and not worsened by the experiment
Chargeback rate (30d)	Card disputes per 1,000 first-30-day card transactions	—	context
Fraud-loss BRL per activated account	Confirmed loss ÷ activated accounts	—	context

14.6 Support & operating cost

KPI	Definition	Benchmark	Target
Onboarding tickets per 1,000 signups	Onboarding-category tickets ÷ signups × 1,000	60–120	≤ 85
Onboarding contact rate %	Signups with ≥1 onboarding ticket	4%–12%	≤ 8%
KYC-appeal rate %	Rejected users who contest the decision	—	context

14.7 Experiment scorecard (Integrity Monitor + Readout pages)

KPI	Definition
Day-7 Activation by arm (+ lift, 95% CI, p)	Primary metric, §8.3
Daily Day-7 Activation by arm	Novelty-effect diagnostic
Daily allocation by arm + cumulative SRM χ^2 p-value	Integrity monitor, §9 — the break story
SRM χ^2 by <code>assignment_source</code> / region / app version	Root-cause diagnostic
Snapshot-vs-log arm-switch count over time	Contamination diagnostic
Contaminated vs clean user counts	Auditable-rule output, §9.7
KYC rejection rate by arm (+ non-inferiority test)	Non-negotiable guardrail
Flagged-fraud rate by arm (+ non-inferiority test)	Non-negotiable guardrail
Onboarding tickets / 1k by arm	Secondary guardrail
Day-7 Activation effect by channel (+ arm × channel interaction test)	Heterogeneity, §8.6
Re-run power / MDE vs original	§11 — the cost of the deadline

15. Targets and plan

- The plan is built on the **same basis as the actuals** (same activation definition, same KYC-rejection and fraud-window definitions).
- **Plan Day-7 Activation:** rising to ≥ **52%** by window end (the redesign is the lever meant to get it there).
- **Plan active users:** ramp to **60k** by month 21.
- **Plan KYC-submit rate:** ≥ **85%**.
- **Plan KYC rejection rate:** hold ≤ **10%** — a ceiling, not a target to optimise down at the cost of legitimate approvals.
- **Plan flagged-fraud rate:** hold ≤ **1.5%**.
- **Plan onboarding tickets:** ≤ **85 / 1,000 signups**.
- "vs Target %" measures exist per KPI; **KYC rejection rate, flagged-fraud rate, manual- review rate, tickets per 1,000 and TTFT are scored lower-is-better**.

16. Out of scope (explicit)

- A second country, region expansion, multi-currency / FX (covered by VoltEdge).
- Credit / lending — underwriting, limits as a credit decision, interest income. The account here is a checking account + debit card; "limits" are a KYC gate, not credit.
- A **fraud-detection model** — the flagged-fraud rate is a guardrail **input**, produced by the existing risk system; this project does not build or tune it.
- Customer-LTV, contribution-margin or unit-economics modelling — activation is the outcome; downstream revenue (interchange, float) is not modelled.
- Marketing-attribution modelling beyond a **first-touch channel** label; no CAC/ROAS.
- App clickstream / screen-level telemetry beyond the defined onboarding-step events.
- Real-time / streaming data; everything is daily batch.
- Real personal data — every user, account, transaction, KYC decision and fraud signal is synthetic and seed-generated.
- The **true treatment effect** and the **exact production-bug magnitudes** — set in Phase 3, detected blind in Phase 10.

17. Resolved-assumptions log

#	Decision	Resolution
A1	Window length & anchor	21 months, Jan 2025 – Sep 2026; data-gen date 2026-09-10; = LumaBank's life to date.
A2	Near-future tail	The last ~3 weeks (after 2026-09-10) exist only to close the re-run's Day-7 windows; not treated as observed history.
A3	Product	Free digital checking account + Mastercard debit. No credit/lending in frame.
A4	Primary metric	Day-7 Activation = account created and ≥ 1 transaction within 7 days of the randomization timestamp.
A5	Randomization point	Signup start (feature-flag exposure), logged with timestamp + app version.
A6	"Active user"	≥ 1 transaction in the trailing 30 days (platform-health metric, not the experiment metric).
A7	Control vs treatment flow	Control = document upload before account creation; treatment = account created in limited state, KYC deferred to a limits gate.
A8	Unit / split	User-level, 50/50, feature-flag hash.
A9	Original experiment window	6 weeks, 2026-07-06 → 2026-08-16; terminated on day 11 by the SRM ALERT.
A10	Expected sample (original)	~13,500 users, ~6,750/arm.
A11	Guardrails	KYC rejection rate (NI, +1.5 pp bound) and flagged-fraud rate (NI, +0.5 pp bound) = non-negotiable; onboarding tickets / 1k = secondary.
A12	Power / MDE (original)	Control Day-7 $\approx$ 48%; MDE +3 pp; α 0.05, power 0.80; ~4,450/arm required, ~6,750 expected.
A13	Heterogeneity	By acquisition channel, with a formal arm $\times$ channel interaction test.
A14	SRM monitoring	Continuous daily χ^2 (cumulative + trailing-7d + segment splits) with WARN at $p < 0.01$, ALERT at $p < 0.001$; runbook in §9.
A15	The break	Mobile release 2026-07-16 resets the variant cache; two effects (contamination + region/timezone-biased fallback); hotfix 2026-07-22.
A16	The decision	Restart, not truncate or exclude-contaminated (trade-off table in scope §4.4 / Phase 10 block 4).
A17	Re-run window & power	4 weeks, 2026-08-24 → 2026-09-20; ~9,000 users; power/MDE recomputed → MDE $\approx$ 3.0 pp; deadline cost stated.
A18	Currency	BRL only; no FX in scope.
A19	True effect & bug magnitudes	Not decided here — set in ETL (Phase 3), detected blind in Phase 10.

End of Phase 1 deliverable. Next gate: Phase 2 — `docs/02_data_architecture.md` (medallion pipeline, the 6 simulated source systems, the curated star schema, and the data-quality + bug-specific injection catalog).

LumaBank — Data Model

Document status: Phase 4 deliverable. **Depends on:** 02_data_architecture.md , 03_data_dictionary.md (auto-generated). Describes the curated star as it will be imported into the Power BI semantic model (Phase 6).

PT. O star curado: matriz de barramento (fatos × dimensões), grão e medidas de cada fato, atributos de cada dimensão, e as relações físicas para o modelo semântico.

1. Bus matrix (facts × dimensions)

Fact ↓ / Dim →	date	user	acquisition_channel	region	app_release	kyc_outcome	onboarding_step	experiment_arm
fact_variant_assignment	• (assigned)	•			•			•
fact_onboarding_step	• (occurred)	•					•	
fact_kyc_decision	(via decided_ts)	•				•		
fact_activation	• (assigned)	•	•	•				•
fact_transaction_day	•	•						
fact_fraud_event	• (flagged)	•						
fact_support_ticket	• (opened)	•						
fact_srm_daily	• (assign_date)							(arm counts as measures)
fact_targets_month	• (month_key, month grain)							

• = direct relationship; (via ...) = reachable through another table, not a physical relationship. Every fact carries *_date_key (int YYYYMMDD) → dim_date[date_key] except fact_targets_month, which joins on month_key (int YYYYMM) at month grain, and fact_kyc_decision, which carries timestamps but no integer date key (decisions are analysed by latency and by joining to fact_activation's date, not on their own calendar axis).

2. Fact tables — grain & measures

Fact	Grain	Additive measures	Non-additive / semi-additive
fact_variant_assignment	one assignment event (≥2 for contaminated users)	count	arm_key, assignment_source, cache_reset_flag, is_first_assignment, is_effective_assignment
fact_onboarding_step	user × step (first occurrence)	count	step_key, seconds_since_signup, is_reached
fact_kyc_decision	one final KYC decision per user	count	kyc_outcome_key, decision_latency_hours, is_rejection, is_manual_review, doc_type, submitted_after_account
fact_activation	one row per signup — platform-wide (is_experiment_subject flags the randomized ones)	onboarding_tickets_n	day7_activated (primary) , day1_activated, day30_activated, hours_to_first_txn, account_opened_flag, kyc_submitted_flag, kyc_rejected_flag, fraud_flag_30d, arm_key, experiment_cohort,

Fact	Grain	Additive measures	Non-additive / semi-additive
			is_experiment_subject , contaminated_flag , assignment_switch_flag , in_analysis_flag
fact_transaction_day	user × local day	txn_count , txn_amount_br1 , pix_count , card_count , boleto_count , transfer_count	is_first_txn_day
fact_fraud_event	one fraud signal	count	signal_type , score , is_confirmed , days_since_activation
fact_support_ticket	one ticket	count	category , channel , is_onboarding
fact_srm_daily	experiment × calendar day (44 rows)	n_control , n_treatment , n_fallback , n_primary	srm_p_cumulative , srm_p_trailing7 , srm_chisq_cumulative , srm_chisq_trailing7 , snapshot_log_mismatch_n , multi_assignment_n , alert_state
fact_targets_month	month × kpi	target_value	kpi

fact_activation is the primary read-out surface: Day-7 Activation, every guardrail, and in_analysis_flag are pre-computed so the report needs almost no DAX and Phase 10 goes straight to the statistics. fact_srm_daily is the integrity-monitor surface: the whole "Experiment Integrity Monitor" page reads from this one table plus dim_date .

3. Dimensions — key attributes

Dimension	Key	Attributes used in the report
dim_date	date_key (int YYYYMMDD)	year , quarter , month , month_name , iso_week , day_of_week , is_weekend , is_holiday_br , is_school_break , season_tag , experiment_phase (pre / original / washout / rerun / post), experiment_day , is_post_deploy
dim_user	user_key	signup_date_key , signup_cohort_week , channel_key , acquisition_channel , region_key , device_os , app_version_at_signup , experiment_arm , experiment_cohort , is_experiment_subject , contaminated_flag , assignment_switch_flag , in_analysis_flag
dim_acquisition_channel	channel_key	channel_name (paid_social / organic / app_store_search / referral / influencer / unknown), channel_group (paid / owned / earned), is_paid
dim_region	region_key	uf (27), uf_name , macro_region , timezone , utc_offset_hours , is_amazon_tz — the axis the post-deploy fallback bias correlates with; -1 = unknown
dim_app_release	app_release_key	version , platform , released_at , rollout_pct , release_type , is_cache_reset_release (true only for the 2026-07-16 row); -1 = unknown
dim_kyc_outcome	kyc_outcome_key	outcome (approved / manual_review / rejected), reason_code , reason_group , is_rejection ; -1 = none
dim_onboarding_step	step_key	step_name (signup_started → personal_data → kyc_upload → account_created → first_transaction → limits_lifted), step_order , funnel_stage
dim_experiment_arm	arm_key	control / treatment / ambiguous — disconnected slicer helper

Sentinel keys. user_key = -1 for any event whose user_id is absent from dim_user (a queue-lag orphan in KYC or fraud decisions). Facts are never dropped for a missing dimension row.

4. Relationships

From (fact)	Column	To	Column	Card.	Cross-filter	Active
fact_variant_assignment	assigned_date_key	dim_date	date_key	* : 1	single	yes

From (fact)	Column	To	Column	Card.	Cross-filter	Active
fact_variant_assignment	user_key	dim_user	user_key	*:1	single	yes
fact_variant_assignment	app_release_key	dim_app_release	app_release_key	*:1	single	yes
fact_onboarding_step	occurred_date_key	dim_date	date_key	*:1	single	yes
fact_onboarding_step	user_key	dim_user	user_key	*:1	single	yes
fact_onboarding_step	step_key	dim_onboarding_step	step_key	*:1	single	yes
fact_kyc_decision	user_key	dim_user	user_key	*:1	single	yes
fact_kyc_decision	kyc_outcome_key	dim_kyc_outcome	kyc_outcome_key	*:1	single	yes
fact_activation	assigned_date_key	dim_date	date_key	*:1	single	yes
fact_activation	user_key	dim_user	user_key	1:1	single	yes
fact_activation	channel_key	dim_acquisition_channel	channel_key	*:1	single	yes
fact_activation	region_key	dim_region	region_key	*:1	single	yes
fact_transaction_day	date_key	dim_date	date_key	*:1	single	yes
fact_transaction_day	user_key	dim_user	user_key	*:1	single	yes
fact_fraud_event	flagged_date_key	dim_date	date_key	*:1	single	yes
fact_fraud_event	user_key	dim_user	user_key	*:1	single	yes
fact_support_ticket	opened_date_key	dim_date	date_key	*:1	single	yes
fact_support_ticket	user_key	dim_user	user_key	*:1	single	yes
fact_srm_daily	assign_date_key	dim_date	date_key	*:1	single	yes
fact_targets_month	month_key	dim_date	month_key (not the PK)	*:*	—	no physical relationship — joined in DAX via TREATAS against dim_date[month_key]

fact_activation is 1:1 with dim_user restricted to is_experiment_subject = true — every randomized user has exactly one row (§ dq_checks). dim_experiment_arm is a disconnected table: the report's arm slicer writes to a measure that filters fact_activation[arm_key] / fact_srm_daily[n_control|n_treatment] directly, not through a relationship — this is what lets one page show *both* arms side by side without a row-splitting filter.

```

erDiagram
    dim_date ||--o{ fact_variant_assignment : "assigned"
    dim_user ||--o{ fact_variant_assignment : ""
    dim_app_release ||--o{ fact_variant_assignment : ""
    dim_date ||--o{ fact_onboarding_step : "occurred"
    dim_user ||--o{ fact_onboarding_step : ""
    dim_onboarding_step ||--o{ fact_onboarding_step : ""
    dim_user ||--o{ fact_kyc_decision : ""
    dim_kyc_outcome ||--o{ fact_kyc_decision : ""
    dim_date ||--o{ fact_activation : "assigned"
    dim_user ||--|| fact_activation : "1:1 (subjects)"
    dim_acquisition_channel ||--o{ fact_activation : ""
    dim_region ||--o{ fact_activation : ""
    dim_date ||--o{ fact_transaction_day : ""
    dim_user ||--o{ fact_transaction_day : ""
    dim_date ||--o{ fact_fraud_event : "flagged"
    dim_user ||--o{ fact_fraud_event : ""
    dim_date ||--o{ fact_support_ticket : "opened"
    dim_user ||--o{ fact_support_ticket : ""
    dim_date ||--o{ fact_srm_daily : "assign_date"

```

5. Measure groups (Phase 7 preview)

Activation & Funnel (Day-7/1/30 Activation, TTFT median/p90, funnel step-reach and conversion by flow) · **Acquisition** (signups and Day-7 Activation by channel) · **Risk & Guardrails** (KYC rejection rate, manual-review rate, flagged-fraud rate, onboarding tickets/1k — all with non-inferiority framing by arm) · **Experiment Integrity** (daily allocation by arm, cumulative + trailing-7d SRM p-value, `alert_state`, snapshot/log mismatch, contaminated-vs-clean counts) · **Experiment Readout** (Day-7 Activation by arm + lift + CI + p, guardrail deltas + NI test, heterogeneity by channel + interaction test, power/MDE context) · **Platform Health** (active users, transaction volume, trend) · **Time Intelligence** (WoW, trailing periods) · **Targets vs Plan**.

End of Phase 4 deliverable (Data Model).